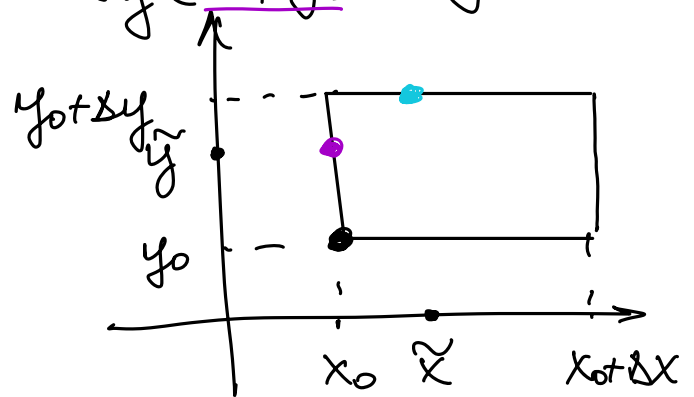


$z = f(x, y)$ $\rho \in \mathbb{R}$ $\exists \delta > 0$
 $\forall (x, y) \in D_f$ $\exists \delta > 0$

$$\Delta f(x_0, y_0) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) =$$

$$= \underbrace{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)}_{[x_0, x_0 + \Delta x] \rightarrow \exists \xi \in D_f} + \underbrace{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}_{[y_0, y_0 + \Delta y] \rightarrow \exists \tilde{y} \in D_f}$$

$$= f'_x(\tilde{x}, y_0 + \Delta y) \cdot \Delta x + f'_y(x_0, \tilde{y}) \cdot \Delta y =$$



$$= (f'_x(x_0, y_0) + \alpha) \Delta x +$$

$$+ (f'_y(x_0, y_0) + \beta) \Delta y =$$

$$\alpha, \beta \xrightarrow{\Delta x, \Delta y \rightarrow 0} 0$$

$$= f'_x(x_0, y_0) \cdot \Delta x + f'_y(x_0, y_0) \Delta y + \alpha \cdot \Delta x + \beta \Delta y$$

$$y = f(x)$$

$$\left(\frac{f'(x)}{a} \right) \Leftrightarrow (a \rightarrow f'(x))$$

$$\Downarrow$$

$$(a \rightarrow f'(x))$$

$$(x_0, y_0) \rightarrow \text{ה'צ"ח } f_y - f_x$$

3 צדען $\Downarrow$

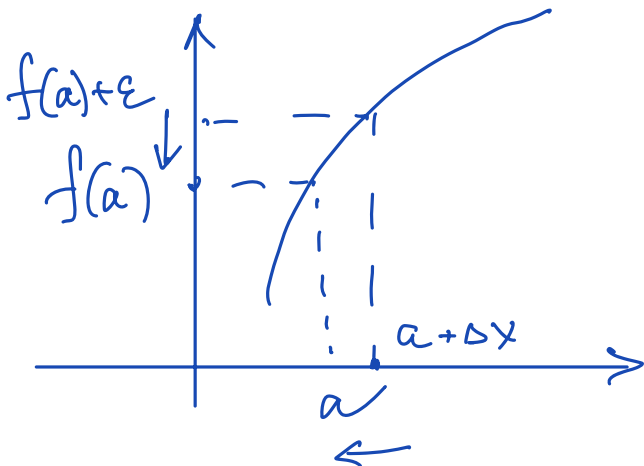
$$(x_0, y_0) \rightarrow \text{ה'צ"ח } f$$

1 צדען $\Downarrow$

$$(x_0, y_0) \rightarrow \text{ה'צ"ח } f$$

2 צדען $\Downarrow$

$$(x_0, y_0) \rightarrow \text{ה'צ"ח } \underline{f'_y} - \underline{f'_x}$$



ה'צ"ח ה'צ"ח ה'צ"ח 1 צדען

$$(1A) \rightarrow \text{ה'צ"ח } \underline{f(x, y) = x^2 - xy} \quad e$$

$$\Delta f(1, 4) = f(1 + \Delta x, 4 + \Delta y) - f(1, 4) =$$

$$= [(1 + \Delta x)^2 - (1 + \Delta x)(4 + \Delta y)] - [1 - 4] =$$

$$= \cancel{1} + \underline{2\Delta x} + (\Delta x)^2 - \cancel{4} - \underline{4\Delta x} - \underline{\Delta y} - \Delta x \Delta y + \cancel{3} =$$

$$= \underbrace{-2\Delta x - \Delta y}_{df(1,4)} + \underbrace{\Delta x \cdot \Delta x}_{\alpha} - \underbrace{\Delta x \cdot \Delta y}_{\beta}$$

$$f'_x(1,4) = (2x-y)|_{(1,4)} = 2-4 = -2$$

$$f'_y(1,4) = -x|_{(1,4)} = -1$$

נתונה הפונקציה $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ 2 נק' ב-3

$(0,0)$ - נק' קריטית

$$f'_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta x f(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x \cdot 0}{(\Delta x)^2 + 0} - 0}{\Delta x} = 0$$

$$f'_y(0,0) = 0$$

$(0,0)$ - נק' צ'ינג'ר f ק'ר

$$f(x,y) = \frac{xy}{x^2+y^2} \xrightarrow{x=y \rightarrow 0} \frac{1}{2} \neq 0$$

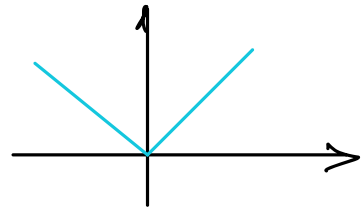
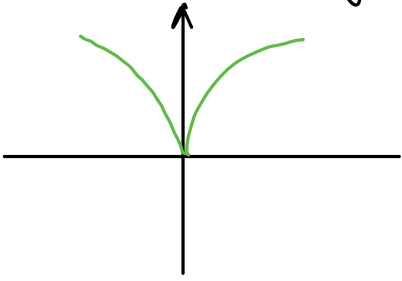
הנק' $(0,0)$ היא נק' צ'ינג'ר f ק'ר $(0,0)$ - נק' צ'ינג'ר

$(0,0)$ - נק' צ'ינג'ר

נתונה $f(x,y) = |x| + \sqrt[3]{y^2}$ 3 נק' ב-3

הנק' $(0,0)$ היא נק' צ'ינג'ר f ק'ר $(0,0)$ - נק' צ'ינג'ר

$$f(0,y) = \sqrt[3]{y^2} \quad \begin{matrix} \text{אם } y \geq 0 \\ y=0 \end{matrix} \quad x=0 \quad \text{אם } x \geq 0 \quad f(x,0) = |x|$$



! אס ? (0,0) - א'3 f(x,y)

$$f(x,y) = \begin{cases} (x^2+y^2) \sin \frac{1}{x^2+y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

ה'3 תי'3 4 הנ'3

ה'3 (0,0) - א'3 (0,0) - א'3

ה'3 f - א'3 (0,0) - א'3

$$x = \rho \cos \theta, \quad y = \rho \sin \theta$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{\rho \rightarrow 0} \rho^2 \sin \frac{1}{\rho^2} = 0$$

0 ≤ 1

ה'3 תי'3 הנ'3

$$f'_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta_x f(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(\Delta x)^2 \sin \frac{1}{(\Delta x)^2} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \Delta x \cdot \sin \frac{1}{(\Delta x)^2} = 0$$

$$f'_y(0,0) = 0 \quad D'f \subset \mathbb{R}' \times \mathbb{R} \times \mathbb{R}$$

$$(x,y) \neq (0,0) \quad f'_x(x,y) \quad \text{AC} \quad \text{C3N1}$$

$$f'_x(x,y) = 2x \cdot \sin \frac{1}{x^2+y^2} + \underbrace{(x^2+y^2)} \cdot \cos \frac{1}{x^2+y^2} \cdot \underbrace{\left(-\frac{1}{(x^2+y^2)^2}\right)}_{2x}$$

$$= \underbrace{2x \sin \frac{1}{x^2+y^2}}_{xy \rightarrow 0} - \underbrace{\frac{2x}{x^2+y^2}}_{y=0, x \rightarrow 0} \underbrace{\cos \frac{1}{x^2+y^2}}_{-1 \leq \cdot \leq 1} \rightarrow 0$$

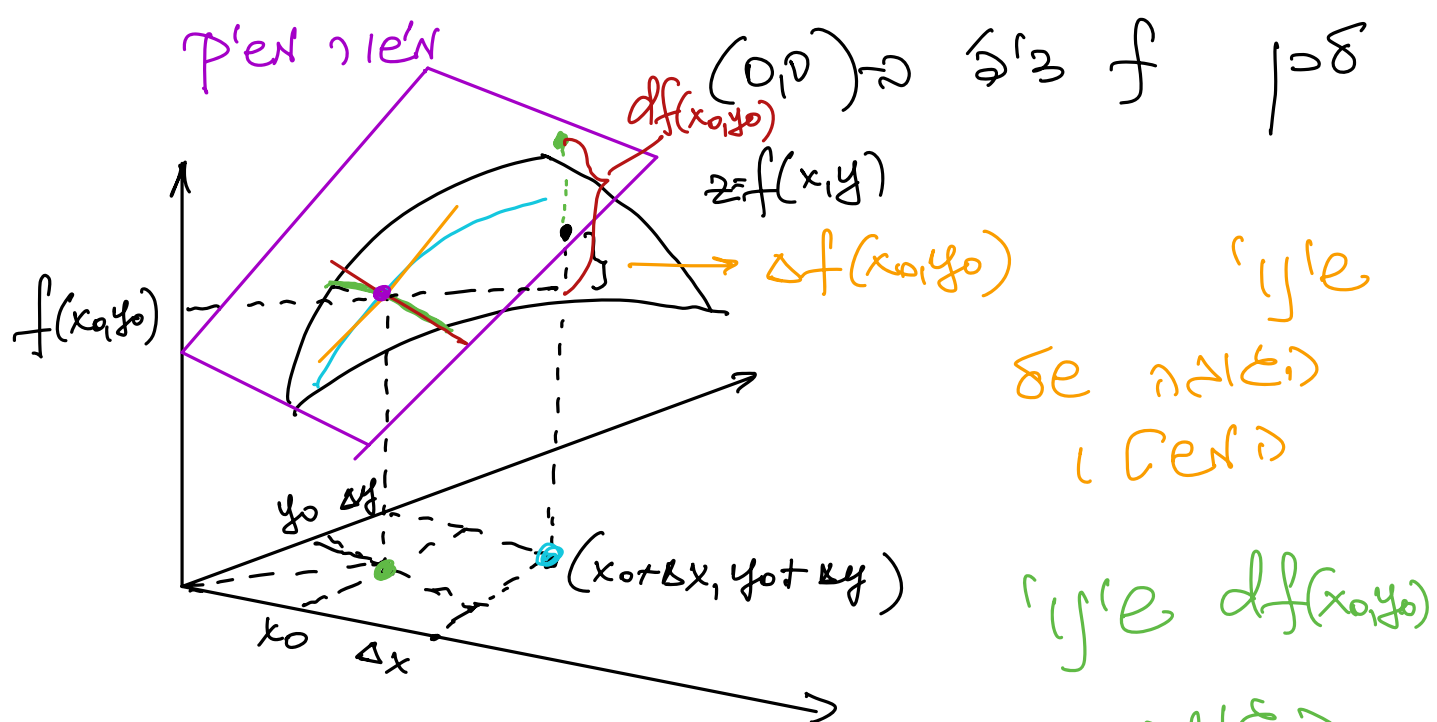
$\lim_{x \rightarrow 0} f(x)$ $\lim_{y \rightarrow 0} f(y)$ $\lim_{x \rightarrow 0} f(x)$ $\lim_{y \rightarrow 0} f(y)$ $\lim_{x \rightarrow 0} f(x)$ $\lim_{y \rightarrow 0} f(y)$

$$\Delta f(\overset{0}{\cancel{x_0}}, \overset{0}{\cancel{y_0}}) = f'_x(0,0) \cdot \Delta x + f'_y(0,0) \cdot \Delta y + \varepsilon \cdot \Delta p$$

$$\varepsilon = \frac{\overbrace{f(0+\Delta x, 0+\Delta y) - f(0,0)}^{\Delta f(0,0)} - f'_x(0,0)\Delta x - f'_y(0,0)\Delta y}{\Delta p} \xrightarrow[\Delta p \rightarrow 0]{\varepsilon \rightarrow 0} 0$$

$$\varepsilon = \frac{[(\Delta x)^2 + (\Delta y)^2] \sin \frac{1}{(\Delta x)^2 + (\Delta y)^2} - 0 - 0 \cdot \Delta x - 0 \cdot \Delta y}{\Delta \rho}$$

$$= \frac{(\Delta p)^2 \sin \frac{1}{(\Delta p)^2}}{\Delta p} = \underbrace{\Delta p}_{\downarrow 0} \cdot \underbrace{\sin \frac{1}{(\Delta p)^2}}_{-1 \leq \sin \frac{1}{(\Delta p)^2} \leq 1} \xrightarrow{\Delta p \rightarrow 0} 0$$



'y'e
 δε ηλθε
 1.000

'y'e $df(x_0, y_0)$

ηλθε

π'εν δε

$$\Delta f(x_0, y_0) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

$$df(x_0, y_0) = f'_x(x_0, y_0) \cdot \Delta x + f'_y(x_0, y_0) \cdot \Delta y$$

$$z - f(x_0, y_0) = f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)$$

$$z - z_0 = A \cdot (x - x_0) + B(y - y_0)$$

ΑΡΗΕΝ
 ηΕΝ
 π'ΕΝ